



Saint Columban College
College of Computing Studies
Pagadian City



What's a Net Cat?

what's a net cat? 



Easy

General Skills

picoCTF 2019

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Description

Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.

Its current status is:

NOT_RUNNING

Launch Instance

Hints 

1

133,745 users solved



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Author: The Analyst: Hyposelenia

Challenge: What's a Net Cat?

Category: General Skills

Date: 01/13/26



I. Objective

. The objective of this task was to successfully connect to a remote server using the **Netcat (nc)** tool and retrieve the flag that is sent by the server upon connection.

II. Background

In cybersecurity, computers often communicate with each other over a **network**. Some challenges do not require downloading files; instead, they require **connecting to another computer** and reading the data it sends back.

Netcat is a simple but powerful tool that allows users to:

- Connect to other computers
- Send and receive text
- Test network services

This challenge introduces the basic idea of **network connections** using a hostname and a port.

III. Tool Used

Oracle VirtualBox (Kali Linux) – used as the **Linux environment** required to run the challenge

- **Linux Terminal** – used to execute **networking commands**
- **nc (netcat)** – a **networking utility** used to connect to a remote server and receive data
- **Hostname** – the name or address of the remote server
- **Port** – a numbered “door” on the server where a service is listening



IV. Methodology

1. Kali Linux was launched using Oracle VirtualBox.
2. The **hostname** and **port number provided by the challenge were noted.**
3. The terminal was opened in Kali Linux.
4. A connection to the remote server was made using the command:
nc <hostname> <port>
5. Once connected, the server immediately sent a message containing the flag.

V. Result

picoCTF{nEtCat_Mast3ry_5BBB6400}

```
(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles]
$ nc fickle-tempest.picocTF.net 52416
You're on your way to becoming the net cat master
picoCTF{nEtCat_Mast3ry_5BBB6400}

(kali㉿kali)-[/media/sf_Downloaded_Cybersecfiles]
$
```



VI. Explanation

Think of the internet like sending messages through pipes.

- A **hostname** is like a house address.
- A **port** is like a specific door in that house.
- **Netcat (nc)** is like knocking on that door and listening.

When the command

nc <hostname> <port>

was used, the computer connected to the challenge's server at the correct address and door.

The server was already programmed to send the flag to anyone who connects, so once the connection was successful, the flag was displayed immediately.

No files were downloaded, and no extra commands were needed — the challenge was about making the connection correctly.

VII. Conclusion

This challenge demonstrates how simple networking tools like **Netcat** can be used to communicate with remote servers. It shows that not all cybersecurity challenges involve hacking or decoding files — sometimes the solution is simply **connecting correctly and reading the response**.

Understanding Netcat is important because it is often used in:

- Networking
- Debugging
- Capture-the-Flag (CTF) challenges
- Basic penetration testing

— **The Analyst: Hyposelenia**